**In this question you must show all stages of your working.**

**Solutions based entirely on calculator technology are not acceptable.**

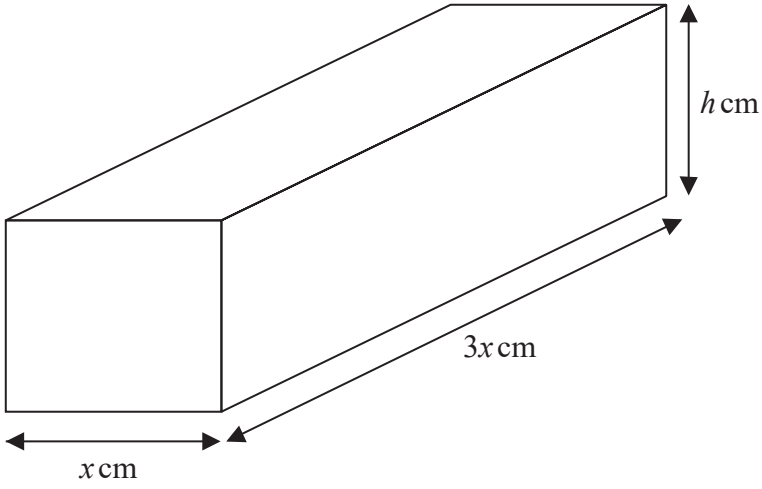



**Figure 2**

A brick is in the shape of a cuboid with width $x$ cm, length $3x$ cm and height $h$ cm, as shown in Figure 2.

The volume of the brick is $972\,\text{cm}^3$

(a) Show that the surface area of the brick, $S\,\text{cm}^2$, is given by

$$S = 6x^2 + \frac{2592}{x}$$

**(3)**

(b) Find $\dfrac{\mathrm{d}S}{\mathrm{d}x}$

**(1)**

(c) Hence find the value of $x$ for which $S$ is stationary.

**(2)**

(d) Find $\dfrac{\mathrm{d}^2S}{\mathrm{d}x^2}$ and hence show that the value of $x$ found in part (c) gives the minimum value of $S$.

**(2)**

(e) Hence find the minimum surface area of the brick.

**(1)**